

The configuration the extrapolated anchor chooses: full results of the inference-configuration sweep

17 August 2026. Last week's multi-day sweep scored 648 combinations of model state and sampling depth across eight research Reynolds numbers, using only the spectral anchor extrapolated from coarse observations, and audited 96 of those combinations against held-out ground truth. This report mines that sweep with information that did not exist when it ran - the measured per-regime anchors built since - and answers three questions: which configuration the extrapolated anchor alone selects at every regime, how accurate that anchor actually was, and whether the structural guard in the selection rule earned its place. The selection replay used here reproduces all sixteen recorded picks of the original run exactly before any variant is computed, so every counterfactual below shares the original's exact semantics.

1. What was swept and how selection works

Candidates at each regime: the untuned base model and the borrowed Reynolds-1000 fine-tune's two reference checkpoints (all regimes), plus twelve natively trained checkpoints at Reynolds 2000 - each run at nine sampling depths, from a single shallow pass to a seven-pass cascade. Every combination was scored blind: the ratio of its output energy to the extrapolated anchor's over wavenumbers 10 to 95, on the anchor's own source sequences. The rule then shortlists cells whose blind score falls in the engraved band, discards those whose fine-scale structure has drifted far from the base reconstruction's (the guard), and takes the survivor nearest the set-point, ties resolved toward shallower depth and less training. Where no cell survives, the fallback takes the cell nearest the set-point regardless of band. Afterwards - and only afterwards - graded retention against ground truth was measured on held-out sequences for the audited subset.

2. The answer: what the extrapolated anchor alone chooses

The table below is the centrepiece: at every regime, the configuration selected using nothing but the extrapolated anchor's blind score (no guard), and its graded quality. Where the two calibration arms of the original ablation disagree, both are shown.

Regime	Chosen by the extrapolated anchor alone	blind	graded retention
1500	three-pass cascade (86 steps) borrowed Re=1000 fine-tune, checkpoint 149	0.801	0.715
2000	three-pass cascade (86 steps) native Re=2000 fine-tune, checkpoint 149	0.789	0.817
3000	single-regime arm: four-pass cascade (110 steps) borrowed Re=1000 fine-tune, checkpoint 149	0.770	0.716
	both-regimes arm: five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 449	0.838	1.133
4000	five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 449	0.805	0.921
5000	five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 449	0.801	0.821
6000	five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 149	0.796	0.736
7000	five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 149	0.789	0.672
8000	five-pass cascade (140 steps) borrowed Re=1000 fine-tune, checkpoint 149	0.793	0.621

In plain terms: near the training regime the anchor chooses a three-pass cascade; by Reynolds 4000 it has moved to the five-pass cascade and stays there through 8000, switching from the later borrowed checkpoint (449) to the earlier, more conservative one (149) as the regime grows more remote. At Reynolds 2000, where native checkpoints competed, it passes over the borrowed models for the native early checkpoint. Graded quality of these blind choices: 0.72 to 0.92 retention through Reynolds 5000, decaying to 0.62 at 8000 - two to four times the best the untuned base achieves anywhere.

3. The guard almost never mattered - the labels did

The original run reported 'no cell in band' at every regime except Reynolds 2000, which read as the guard vetoing everything. The faithful replay resolves this: with the guard removed, the selected cell is IDENTICAL at fifteen of the sixteen regime-and-arm combinations. The guard's vetoes changed the label (in-band pick versus fallback), but the fallback landed on the same cell the blind score would have chosen anyway. The single exception is Reynolds 1500 in the single-regime arm, where the guarded rule's fallback chose checkpoint 449 (graded 0.745) while the pure blind rule chooses checkpoint 149 (graded 0.715) - a 0.03 difference, in the guard's favour. Conclusion: the guard is close to a no-op in outcomes and its reference drifts across regimes; a revised rule can drop it or re-specify it without loss.

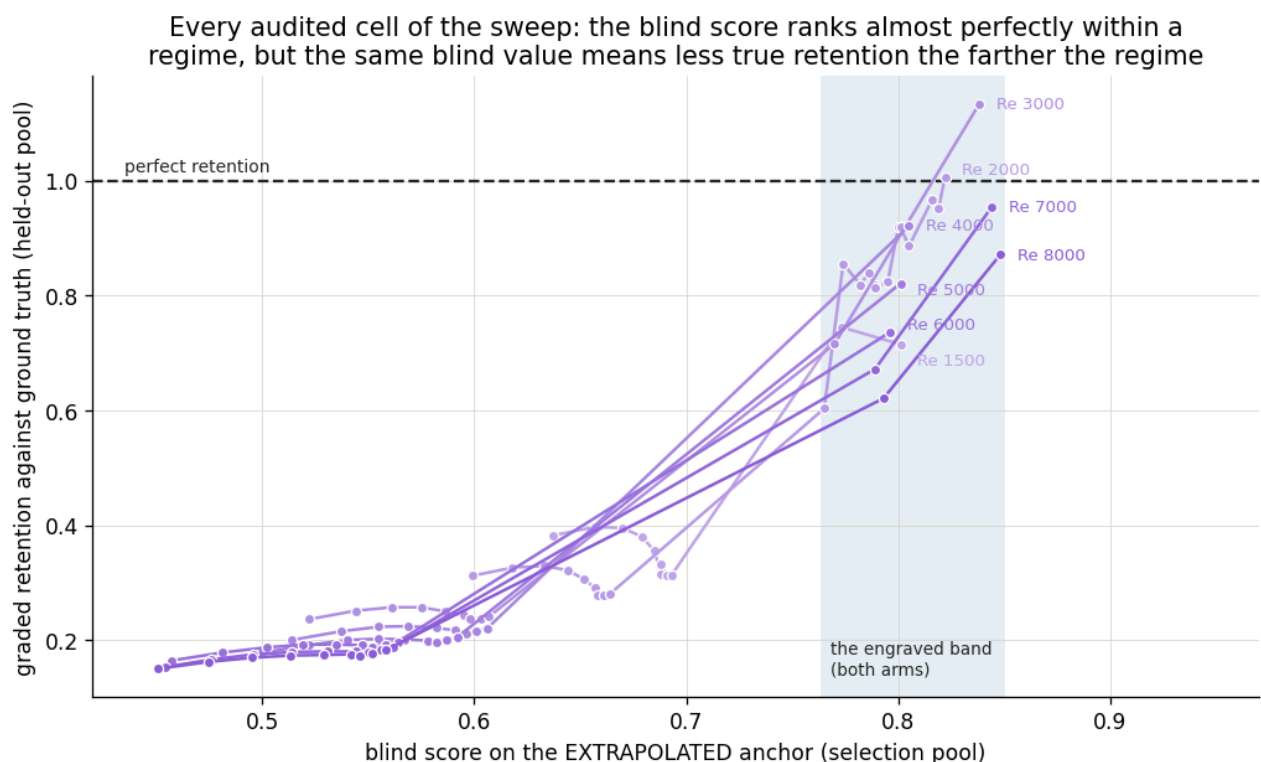
4. How accurate was the extrapolated anchor, exactly

Comparing the extrapolated anchor against the measured anchor (built later from training-sequence ground truth) band by band shows the anchor's error is a SHAPE error, not a level error: it is too energetic in the middle wavenumbers and too weak in the tail, at every regime.

Regime	blind band [10,96): extrapolated / measured	reward band [32,96)
1500	1.074 (hot)	0.934 (cold)
2000	1.109 (hot)	0.935 (cold)
3000	1.186 (hot)	0.964 (cold)
4000	1.218 (hot)	0.964 (cold)
5000	1.125 (hot)	0.885 (cold)
6000	1.221 (hot)	0.949 (cold)
7000	1.183 (hot)	0.887 (cold)
8000	1.141 (hot)	0.848 (cold)

Hot by 7 to 22 percent where the blind score integrates, cold by 4 to 15 percent where the reward concentrates - the 'too narrow in k' diagnosis of the earlier audit, now quantified per regime. These ratios are directly usable as a correction curve for a second-generation anchor.

5. Despite that, the blind score was nearly optimally calibrated



All 96 audited cells: blind score (horizontal) against graded retention (vertical), one line per regime.

Two facts. Within any single regime the blind score ranks configurations almost perfectly (correlation 0.85 to 0.97; pooled 0.93): whatever the anchor's absolute error, MORE blind energy reliably means MORE true energy. And at the two regimes where audited models actually cross perfect retention, the blind value corresponding to retention exactly 1 is 0.821 (Reynolds 2000) and 0.816 (Reynolds 3000) - the engraved set-point of 0.81 was within one percent of optimal, twice, independently. What does NOT transfer is the absolute meaning of a blind value across regimes: inside the engraved band, true retention fans out from 1.13 at Reynolds 3000 to 0.62 at 8000. Most of that fan-out is model reach, not anchor error - at the far regimes the audited models never cross

retention 1 at ANY depth, so no blind value could map to perfection there.

6. What this supports going forward

(1) The engraved blind machinery stands: selection from the extrapolated anchor alone reproduces the full rule's choices at 15 of 16 arm-regime combinations and needed neither the guard nor the second calibration regime. (2) A second-generation anchor should correct the k-shape using the per-regime hot/cold ratios of section 4; against corrected anchors the set-point re-expresses to roughly 0.93. (3) The far-regime ceiling is a model limitation, not a selection limitation - the borrowed checkpoint simply runs out of reach past Reynolds 5000, which is the same conclusion the conditioning arc reached from the other direction. Provenance: sweep `src/ddpo_ft/setpoint_ablation.py` (results `base_results/setpoint_ablation.npz`); replay `src/ddpo_ft/ablation_choice_replay.py` (validated against the run log); anchor comparison `src/ddpo_ft/ablation_mining.py`.